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ScienceDirect
Mathematical and Computer Modelling 46 (2007) 138–150
MATHEMATICAL AND COMPUTER MODELLING
www.elsevier.com/locate/mcm

Testing the capital asset pricing model with Local Maximum Likelihood methods

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Received 16 May 2006; accepted 15 December 2006

Abstract

This paper follows the approach of Wang [K. Wang, Asset pricing with conditioning information: A new test, Journal of Finance 58 (2003) 161–196] in order to test the conditional version of Sharpe–Lintner CAPM by adopting Local Maximum Likelihood nonparametric methods. This methodology does not only avoid the misspecification of betas, risk premiums and the stochastic discount factor but is also expected to perform better when compared with other more traditional methods such as the constant Nadaraya–Watson kernel estimator due to its superior performance at the sample extremes.
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Keywords: Capital asset pricing model; Local maximum likelihood

1. Introduction

Asset pricing models have attracted considerable attention in the finance literature. Various models such as Arbitrage Pricing Theory (APT) or Consumption-based General Equilibrium models have been developed in the finance literature to forecast asset returns; however, none of these models have been as popular as the capital asset pricing model (CAPM) of Sharpe [29] and Lintner [27]. The central assumption of CAPM is that a market portfolio of invested wealth is mean–variance efficient and under this assumption it predicts that the risk premiums on individual assets will be proportional to the risk premium of the market portfolio and the beta coefficient of the market security. The latter measures the sensitivity of the market security's return to market fluctuations. This simple version of CAPM is also known as unconditional or static CAPM due to the fact that the relation between individual securities and market portfolio, as implied by the betas, is assumed to be time-invariant and stable in this framework.

The CAPM has been examined and tested by a great number of authors during the past decades and various anomalies of the static CAPM have been documented as a result of these studies. Papers by Fama and French [6,8, 9] have found that the static CAPM does not hold empirically. The empirical results against the unconditional CAPM were so strong that it led some authors to conclude that CAPM is dead. However, there were also numerous attempts to explore the nature of these anomalies and correct them. Some of these attempts concentrated on testing the stability of the betas. Following Levy's [26] suggestion to compute beta's for separate markets, Fabozzi and Francis [5] estimated

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doi:10.1016/j.mcm.2006.12.014

and tested the stability of beta's over bull and bear markets. They found out that single index market models of CAPM were unaffected by bull and bear market conditions.

In another context, studies by Keim and Stambaugh [25] and Breen, Glosten and Jagannathan [1] showed that betas in the CAPM framework are not time-invariant and they vary during the business cycles as shown by Chen [2] and Ferson and Harvey [13]. In their paper Jagannathan and Wang [21] have developed a conditional version of CAPM where the betas of securities are conditioned on the information set available to investors at a particular point in time and are subject to change with the fluctuating economic conditions over time. Dybvig and Ross [4] and Hansen and Richard [18] demonstrated that even though the static CAPM may result in serious pricing errors, the conditional CAPM might still hold.

The development of the conditional CAPM motivated a new literature that focused on the formation and testing of these models. Despite offering solutions to some of the static version's anomalies, the conditional CAPM also introduced some new problems into the literature. One such problem was the choice of conditioning variables and the lack of theory in how to form the relationship between the betas and conditioning variables. At first, the conditional CAPM that exhibited betas as linear functions of conditioning variables was adopted by He et al. [20]. However, the performance of such specifications was sometimes even worse than the static CAPM of Sharpe [29] and Lintner [27]. It has been argued that the validity of such tests should not be trusted, since their results are strongly influenced by the modelling assumptions inherent in them. In fact, Ghysels [17] showed that when beta dynamics are misspecified, the pricing errors of the conditional CAPM are much likely to be greater than those of the unconditional version.

The above findings motivated researchers to focus on nonparametric estimation techniques in an attempt to avoid relying on a specific functional form in the estimation of betas. Wang [31] adopted such a nonparametric approach for specifying the beta risk dynamics by using the Nadaraya–Watson kernel estimator within a stochastic discount factor (SDF) framework to test the restrictions implied by the conditional CAPM. In his paper, he showed that conditional CAPM is superior to the unconditional version despite the fact that pricing errors committed by the conditional CAPM are still statistically significant even within this nonparametric setup.

In this paper, we adopt a similar testing methodology as in Wang [31], using instead Local Maximum Likelihood as the nonparametric estimation method. It is known that the Nadaraya–Watson Kernel Method performs poorly in tail estimation (also known as boundary effect) and it is a stylized fact that stock returns exhibit fatter tails. Our goal is to improve the explanatory power of the conditional CAPM model via adopting a more precise estimation technique and compare our results with those of Wang [31] to see whether the results he obtained are robust to different local smoothing techniques.

The paper proceeds as follows. The nonparametric presentation of conditional CAPM and the testing methodology are presented in the next section. The Local Maximum Likelihood technique is briefly explained in section three. Section four provides the data description and empirical results. Finally we conclude.

2. Conditional capital asset pricing model (CAPM) and testing methodology

The so called static or unconditional CAPM is depicted by the following equation:

$$[E(r_i) - r_f] = \beta_i [E(r_m) - r_f] \quad \text{where } \beta_i = \frac{\mathrm{Cov}(r_i, r_m)}{\sigma_m^2}. \tag{2.1}$$

In words, Eq. (2.1) states that the risk premium on individual assets will be proportional to the risk premium on the market portfolio and the beta coefficient of the individual asset that shows the sensitivity of the individual asset's return to the changes in the market return. Therefore, as the asset gets riskier, i.e. the more sensitive it is to the market changes, investors should be compensated with higher returns in order to invest in that asset.

The paper by Fama and French [6] added fuel to a still continuing debate on whether the beta predicted by the static CAPM is an adequate measurement of risk or not. The studies conducted on the defense of CAPM focused mainly on either using improved econometric techniques in an attempt to improve the accuracy of CAPM estimates by eliminating the noise; or on developing the theoretical framework of CAPM via taking lessons from the results of the tests that have been conducted in the literature. The conditional CAPM or the so-called "dynamic" CAPM was introduced in such an attempt by Ferson and Harvey [14] and Jagannathan and Wang [21]. Basically, they used the following time-conditional framework to express the risk dynamics implied by CAPM:

$$E(r_{i,t+1} \mid I_t) = \frac{\mathrm{Cov}(r_{i,t+1}, r_{p,t+1} \mid I_t)}{\mathrm{Var}(r_{p,t+1} \mid I_t)} E(r_{p,t+1} \mid I_t) \quad \text{for } t = 1, \ldots, N \tag{2.2}$$

where N stands for the sample size, $r_{p,t+1}$ stands for the return on the market portfolio in excess of the risk free rate and $r_{i,t+1}$ stands for the excess return on the ith asset. This version of CAPM adopted the same relationship between risk premium on an individual asset and the market portfolio; however, betas and risk premiums are allowed to vary over time depending on the economic conditions in this framework. It represents the information set available to investors at time t and they estimate the expected values of risk premiums and betas conditional on the information they have at time t.

Although this dynamic version of the CAPM was able to improve CAPM's reputation up to a degree, it brought along problems of its own. Ironically, the conditional version of the Fama and French model has been found to fail miserably in the empirical tests conducted by He et al. [20], Ferson and Siegel [16], and Ferson and Harvey [15]. The main problem that occurred with the conditional CAPM was the lack of theoretical support on specifying the relation between conditioning variables and risk premiums as well as betas. In fact, Ghysels [17] showed that the results of such tests are mainly influenced by the specification of betas or the risk premiums which were usually assumed to be linear in such empirical tests. He argued that misspecification of risk dynamics was the major reason behind the failure of conditional CAPM in such tests.

This development induced researchers to develop models that are free from misspecification errors. One potential solution was the adoption of nonparametric techniques in estimation of conditional expectations. Wang [31] used such an approach and suggested the following testing methodology that will also be used in this paper. His inference methodology relies on the restrictions implied by CAPM on the stochastic discount factor (SDF) framework.[1] SDF is a very general framework that is applicable to any asset pricing equation that generates the following common result:

$$E_t(m_{t+1} R_{i,t+1}) = 1 \tag{2.3}$$

where E_t denotes the conditional expectation, m_{t+1} is the SDF and $R_{i,t+1}$ is the gross return on the ith asset. The restrictions on the SDF depend on the model. Eq. (2.3) can be rewritten for excess returns as follows:

$$E_t(m_{t+1} r_{i,t+1}) = 0 \quad \text{for } i = 1, \ldots, n \tag{2.4}$$

where n denotes the number of assets and $r_{i,t+1}$ stands for excess return on ith asset.

The conditional CAPM implies that the return on individual and market portfolios follow the relation depicted by Eq. (2.2) which can be rewritten as:

$$\begin{aligned} E(r_{i,t+1} \mid I_t)\lfloor E(r_{p,t+1}^2 \mid I_t) - [E(r_{p,t+1} \mid I_t)]^2 \rfloor &= E(r_{i,t+1} r_{p,t+1} \mid I_t) E(r_{p,t+1} \mid I_t) \\ &\quad - E(r_{i,t+1} \mid I_t)[E(r_{p,t+1} \mid I_t)]^2 \end{aligned}$$

by using

$$\mathrm{Cov}(r_{i,t+1}, r_{p,t+1} \mid I_t) = E(r_{i,t+1} r_{p,t+1} \mid I_t) - E(r_{i,t+1} \mid I_t) E(r_{p,t+1} \mid I_t)$$

as well as

$$\mathrm{Var}(r_{p,t+1} \mid I_t) = E(r_{p,t+1}^2 \mid I_t) - [E(r_{p,t+1} \mid I_t)]^2.$$

By cancelling the common term, $E(r_{i,t+1} \mid I_t)[E(r_{p,t+1} \mid I_t)]^2$ at both sides we can get the simplified equation:

$$E(r_{i,t+1} \mid I_t) = \frac{E(r_{i,t+1} r_{p,t+1} \mid I_t) E(r_{p,t+1} \mid I_t)}{E(r_{p,t+1}^2 \mid I_t)} E(r_{p,t+1} \mid I_t). \tag{2.5}$$

Eq. (2.5) is also known as the cross-moment representation.

For empirical testing purposes, there would be a set of conditioning variables, x_t, that would be used to characterize the information set I_t:

$$E(r_{p,t+1} \mid I_t) = E(r_{p,t+1} | x_t) \quad \text{and} \quad E(r_{p,t+1}^2 \mid I_t) = E(r_{p,t+1}^2 \mid x_t) \tag{2.6}$$

[1] See Cochrane [3] for a detailed explanation of the SDF framework.

where the excess returns as well as the conditioning variables are assumed to be strictly stationary.

Following Wang [31] we define $g_p(x_t) = E(r_{p,t+1} \mid x_t)$, $g_{pp}(x_t) = E(r^2_{p,t+1} \mid x_t)$ and $b(x_t) = g_p(x_t)/g_{pp}(x_t)$; under the assumption that Eq. (2.6) holds for both representations, the conditional pricing errors of CAPM can be expressed as:

$$E(r_{i,t+1} \mid I_t) - \frac{E(r_{i,t+1} r_{p,t+1} \mid I_t) E(r_{p,t+1} \mid I_t)}{E(r^2_{p,t+1} \mid I_t)} E(r_{p,t+1} \mid I_t) = E(m_{t+1} r_{i,t+1} \mid I_t) = 0 \tag{2.7}$$

where $m_{t+1} = 1 - b(x_t) r_{p,t+1}$ and therefore Eq. (2.2) can be rewritten as Eq. (2.7) that is equivalent to $E_t(m_{t+1} r_{i,t+1}) = 0$ as stated before.

A simple test of the conditional CAPM can be conducted via estimating the following regression and then checking whether the coefficients are significantly different from zero:

$$e_{i,t+1} = z_t' \delta_t + u_{i,t+1} \tag{2.8}$$

$e_{i,t+1} = m_{t+1} r_{i,t+1}$ and z_t is a $qx\ 1$ vector of observed variables in I_t. The moment condition implied by Eq. (2.2) can be stated as $\delta = 0$ where $\delta = (\delta_1' \delta_2' \delta_3' \ldots \delta_n')'$. Wang [31] used the Nadaraya–Watson kernel estimator and Rosenblatt–Parzen kernel density estimator in order to estimate the SDF nonparametrically, $\hat{m}_{t+1}$. Moreover, he adopted a weighted least squares regression to estimate Eq. (2.8) where the weighting function is chosen to be $\hat{w}_t = \hat{f}(x_t) \hat{g}_{pp}(x_t)$ for technical reasons.[2] After getting $\hat{\delta}_N = (\hat{\delta}_1' \ldots \hat{\delta}_n')'$ from the weighted least squares estimator, a joint test of pricing errors on n portfolios, can be conducted via $N \hat{\delta}_N' \hat{\Omega}_N^{-1} \hat{\delta}_N$. This test has a chi-squared distribution with "qn" degrees of freedom, under the null hypothesis that conditional CAPM holds. $\hat{\Omega}_N$ is a consistent estimator of Ω and details about how to calculate $\hat{\Omega}_N$ will be given in the Appendix A. The major advantage of such a test is that it has a standard limiting multivariate normal distribution and it converges as the fast parametric convergence rate despite the fact that variables that are used for the computation of $\hat{\delta}_N$ are obtained with nonparametric estimators.

3. Local Maximum Likelihood method

Wang's [31] rejection of the conditional CAPM of Sharpe–Lintner may be due to the nonparametric smoothing technique that he employed. Our aim is to test whether this is the case by using a local smoothing nonparametric technique that has superior properties to the simple kernel method used by Wang. The commonly used Nadaraya–Watson kernel estimator[3] is known to exhibit major bias in estimating extreme values and it is outperformed by Local Maximum Likelihood (LML), see Jasiak and Gourieroux [22], and Fan and Gu [12]. In this paper we have chosen LML as the nonparametric estimation technique anticipating more robust CAPM estimates and test results. Before proceeding to empirical results, a brief explanation of Local Maximum Likelihood procedure will be given in this section.

Following Fan, Farmen and Gijbels [11] the estimation procedure of LML can be described as follows.

The ith observation, (X_i, Y_i), in the whole sample, $(X_1, Y_1), (X_2, Y_2), \ldots, (X_n, Y_n)$ contributes $\ell[g(X_i, Y_i)]$ to the conditional log-likelihood, where $g(.)$ is an unknown parameterized function of interest, i.e. $g(x) = g_\theta(x)$. The conditional log-likelihood of n observations is given by $\sum_{i=1}^n \ell\{g(X_i, Y_i)\}$, where θ is the unique solution to the log-likelihood equation:

$$E\left\{\ell'\left(g(X_i, Y_i)\right)\right\} = 0 \quad \text{where } \ell'(t, u) = \frac{\partial}{\partial t} \ell(t, u)$$

$g(.)$ is completely unknown so we want to estimate it nonparametrically around a local point: $g(x_0)$. Assume that g has a $(p+1)$th continuous derivative at point x_0. For data points, X_i, in the neighbourhood of x_0 we approximate $g(X_i)$ via Taylor series expansion with a polynomial degree of p

$$g(X_i) \cong g(x_0) + g'(x_0)(X_i - x_0) + \cdots + \frac{g^p(x_0)}{p!}(X_i - x_0)^p = X_i^{\mathrm{T}} \beta_0$$

[2] For details see Wang [31].

[3] See Jones, Davies and Park [23] for details.

where $X_i = (1, (X_i - x_0), (X_i - x_0)^2, \ldots, (X_i - x_0)^p)$

$$\beta_0 = \left(\beta_0^0, \ldots, \beta_0^p\right) \quad \text{and} \quad \beta_0^v = \frac{g^{(v)}(x_0)}{v!} \quad \text{where } v = 0, 1, \ldots, p. \tag{3.1}$$

For data points (X_i, Y_i) in a neighbourhood of X_o, the log-likelihood is $\ell(X_i^{\mathrm{T}}\beta_0, Y_i)$; weighted by $K_h(X_i - x_0)$ where $K_h(X_i - x_0) = K(\frac{X_i - x_0}{h})/h$.
Then the conditional local kernel-weighted log-likelihood is given by:

$$L_p(x_0, h, \beta) = \sum_{i=1}^{n} \ell(X_i^{\mathrm{T}}\beta_0, Y_i) K_h(X_i - x_0). \tag{3.2}$$

Maximizing (3.2) with respect to β yields the vector of estimators $\hat{\beta} = (\hat{\beta}_0, \ldots, \hat{\beta}_p)$
Estimators $\hat{g}^{(v)}(x_0)$ for $g^{(v)}(x_0)$ where $v = 0, 1, \ldots, p$ are given by: $\hat{g}^{(v)}(x_0) = v!\hat{\beta}_v$

Consider the normal regression model $Y = g(X) + \varepsilon$ where $\varepsilon \sim N(0, \sigma^2)$, X and ε are independent. Then the conditional local log-likelihood is given by:

$$-\log\sqrt{2\pi\sigma} \sum_{i=1}^{n} K_h(X_i - x_0) - \frac{1}{2\sigma^2} \sum_{i=1}^{n} \left\{ Y_i - \sum_{j=0}^{p} \beta_j (X_i - x_0)^j \right\}^2 K_h(X_i - x_0). \tag{3.3}$$

This has to be maximized with respect to β and it is equivalent to minimizing:

$$\sum_{i=1}^{n} \left\{ Y_i - \sum_{j=0}^{p} \beta_j (X_i - x_0)^j \right\}^2 K_h(X_i - x_0). \tag{3.4}$$

In the above example, the unknown function was a mean regression function just as it would be in our context. In other words, $g(.)$ will be the conditional expectations of the return of market and individual portfolios as well as their joint conditional expectations.

Note that, even when σ depends on the location x_0, the local kernel weighted method can still be used to estimate $g(.)$ because of local homoscedastcity: $\sigma^2(X_i) \approx \sigma^2(x_0)$ for X_i in a neighbourhood of x_0. One way of estimating $\sigma(.)$ is to estimate β first by regarding $\sigma^2(.)$ locally as a constant and then applying the local modeling idea to $\log\{\sigma^2(.)\}$ by using a different bandwidth, see Ruppert, Wand, Holst and Hossjer [28].

3.1. Bias and variance of the LML estimator and bandwidth selection

The bias and variance of the estimation process are crucial in this method because they are the building blocks of bandwidth selection as well as confidence interval computations. The bias of the $\hat{\beta}$ estimator comes from the approximation error in the Taylor expansion that can be expressed as:

$$r(X_i) = g(X_i) - \sum_{j=0}^{p} g^{(j)}(x_0)(X_i - x_0)^j / j!. \tag{3.5}$$

Let Eq. (3.5) denote the approximation error at the point X_i. Suppose that the $(p + a + 1)$th derivative of the function g exists at the point x_0 for some $a > 0$. Then a further expansion of $g(X_i)$ gives an approximation to approximation error:

$$r(X_i) = \beta_{p+1}^0 (X_i - x_0)^{p+1} + \cdots + \beta_{p+a}^0 (X_i - x_0)^{p+a} \equiv r_i \tag{3.6}$$

where "a" denotes the order of approximation. The choice of "a" will have some effect on the performance of the estimated bias. Fan and Gijbels [10] suggest that $a = 2$ will provide good practical performance while the computational burden will be modest.

If the quantities r_i were known, then a more precise log-likelihood estimation can be obtained via:

$$L_p^*(x_0, h, \beta) = \sum_{i=1}^{n} \ell(X_i^{\mathrm{T}}\beta + r_i, Y_i) K_h(X_i - x_0). \tag{3.7}$$

The maximizer of local log-likelihood $L_p^*(x_0, h, \beta)$ in Eq. (3.7) will be denoted by $\hat{\beta}^* = \hat{\beta}^*(x_0)$. The bias of the $\hat{\beta}(x_0)$ can be estimated by $\hat{\beta}(x_0) - \hat{\beta}^*(x_0)$ as follows.

Let $L_p^{*'}(x_0, h, \beta) = \frac{\partial}{\partial\beta} L_p^*(x_0, h, \beta)$ and $L_p^{*''}(x_0, h, \beta) = \frac{\partial^2}{\partial\beta^2} L_p^*(x_0, h, \beta)$ denote the gradient vector and Hessian matrix of the local log-likelihood L_p^* respectively and since $\hat{\beta}^*(x_0)$ is the maximizer of $L_p^*(x_0, h, \beta)$, a Taylor series expansion gives:

$$0 = L_p^{*'}(x_0, h, \hat{\beta}) \approx L_p^{*'}(x_0, h, \hat{\beta}) + L_p^{*''}(x_0, h, \beta)(\hat{\beta}^*(x_0) - \hat{\beta}(x_0)). \tag{3.8}$$

This would lead us to define the estimated bias vector:

$$\hat{b}_p(x_0) = [L_p^{*''}(x_0, h, \beta)]^{-1} L_p^{*'}(x_0, h, \hat{\beta}). \tag{3.9}$$

To get a better insight into the bias approximation, let us look at the normal likelihood given at Eq. (3.3) before:

$$-\log\sqrt{2\Pi\sigma}\sum_{i=1}^{n} K_h(X_i - x_0) - \frac{1}{2\sigma^2}\sum_{i=1}^{n}\left\{Y_i - \sum_{j=0}^{p}\beta_j(X_i - x_0)^j\right\}^2 K_h(X_i - x_0)$$

X stands for the design matrix of the regression problem, i.e. the $nx(p+1)$ matrix whose (i, j)th element is $(X_i - x_0)^{j-1}$ and let $\mathbf{W} = \mathrm{diag}\{K_h(X_i - x_0)\}$ be the diagonal matrix containing the weights. Except for a constant factor, the local log-likelihood can be written

$$L_p^*(x_0, h, \beta) = (y - X\beta - r)^{\mathrm{T}}\, W\, (y - X\beta - r) \quad \text{where } y = (Y_1, \ldots, Y_n)^{\mathrm{T}} \text{ and } r = (r_1, \ldots, r_n)^{\mathrm{T}}.$$

Furthermore, $\hat{\beta} = (X^{\mathrm{T}}WX)^{-1}X^{\mathrm{T}}Wy$ and hence $L_p^{*'}(x_0, h, \beta) = 2X^{\mathrm{T}}Wr$ and $L_p^{*''}(x_0, h, \beta) = 2X^{\mathrm{T}}WX$. Therefore $\hat{b}_p(x_0) = (X^{\mathrm{T}}WX)^{-1}(X^{\mathrm{T}}Wr)$ which is equal to the approximation of bias $E(\hat{\beta}|X) - \beta^0$.

Recall that approximated bias in (3.9) depends on unknown quantities r_i, which will be estimated by fitting a polynomial of degree $p + a$ locally via Eq. (3.2) using a pilot bandwidth h^*. This gives the estimates $\hat{\beta}^{(p+a)} = (\hat{\beta}_0, \ldots, \hat{\beta}_{p+a})^{\mathrm{T}}$ which are then substituted into expression (3.6) yielding the estimates $\hat{r}_1, \ldots, \hat{r}_n$ of $r_1, \ldots, r_n$. These estimates are then substituted into (3.7) leading to the estimated bias as in (3.9). Denote the estimated bias of $\hat{\beta}_v$ by $\hat{\beta}_{p,v}(x_0, h)$, the $(v+1)$th element of $\hat{b}_p(x_0)$.

By a slight modification of Eq. (3.8) we can rewrite it as follows:

$$(\hat{\beta} - \beta_0) \approx -[L_p''(x_0, h, \beta_0)]^{-1} L_p'(x_0, h, \beta_0).$$

This would give an approximation for the conditional variance:

$$\mathrm{Var}(\hat{\beta}|X) \approx [L_p''(x_0, h, \beta_0)]^{-1}\mathrm{Var}(L_p'(x_0, h, \beta_0))[L_p''(x_0, h, \beta_0)]^{-1}. \tag{3.10}$$

Following Fan, Gijbels and Farmen [11], it can be shown that the sample estimate of the variance for a normal density is given by the following equation,

$$\sum(x_0) = \sigma^2(x_0) S_n^{-1}\bar{S}_n S_n^{-1} \quad \text{where } S_n = X^{\mathrm{T}}WX \tag{3.11}$$

where $\hat{\sigma}^2(x_0) = \frac{\sum_{i=1}^{n}(Y_i - \hat{Y}_i)^2 K_{h^*}(X_i - x_0)}{\sum_{i=1}^{n} K_{h^*}(X_i - x_0)}$ for this special case. $\hat{Y}_i$ in the above equation is obtained from a $(p+a)$th order fit, $\hat{Y}_i = (X_i^*)^{\mathrm{T}}\hat{\beta}^{(p+a)}$.

The crucial part in nonparametric estimation is the bandwidth selection. In LML estimation, both the variance and the bias of the estimator rely on the bandwidth. In order to choose the bandwidth, usually some variant of the Mean Squared Error (MSE) selection criteria is adopted in the literature. Fan and Gijbels [10] follow the same approach and suggest the following criterion called Residual Squares Criterion:

$$\mathrm{RSC}(x_{o;}h) = \hat{\sigma}^2(x_0)\{1 + (p+1)/N\} \tag{3.12}$$

where $1/N$ is the first diagonal element of $(X^{\mathrm{T}}WX)^{-1}(X^{\mathrm{T}}W^2X)(X^{\mathrm{T}}WX)^{-1}$ and $\hat{\sigma}^2(x_0)$ is the normalized weighted sum of squares after fitting locally a pth order polynomial given by:

$$\hat{\sigma}^2(x_0) = \frac{\sum_{i=1}^{n}(Y_i - \hat{Y}_i)^2 K_{h^*}(X_i - x_0)}{\mathrm{tr}(W) - \mathrm{tr}[(X^{\mathrm{T}}WX)^{-1}(X^{\mathrm{T}}W^2X)]}.$$

The logic behind Eq. (3.12) is as follows. When the bandwidth, h, is too small then the variance of the fit would be high which would induce $1/N$ to be large as well. However, when h is too high, then the bias would be high which would induce $\hat{\sigma}^2(x_0)$ to be large as well. Therefore, Eq. (3.12) gives a stable medium that penalizes for extreme values of h. Furthermore, Fan and Gijbels [10] demonstrate that the h that minimizes Eq. (3.12) is a constant factor away from the optimal pilot bandwidth:

$$h_{\mathrm{opt}}(x_0) = \mathrm{adj}_{p,v}(K)h_o(x_0). \tag{3.13}$$

This constant, $\mathrm{adj}_{p,v}(K)$, depends on the choice of kernel, the order of polynomial fit and the moment that is being estimated, and the values that it can take for some popular kernels are given by Fan and Gijbels [10].

The pilot bandwidth is selected with the RSC as explained above with a $(p+a)$th order polynomial fit. Then this optimal pilot bandwidth is used for the estimation of $\hat{\beta}^{(p+a)} = (\hat{\beta}_0, \ldots, \hat{\beta}_{p+a})^{\mathrm{T}}$ estimator which in turn would be used to estimate the bias $\hat{B}_{p,v}(x_0; h)$ and variance of $\hat{V}_{p,v}(x_0; h)$ of $\hat{B}_v$ that are respectively the $(v+1)$th element of Eq. (3.9) and $(v+1)$th diagonal element of Eq. (3.11). Finally, the optimal bandwidth that will be used to estimate the pth order fit would be chosen with the MSE of $\hat{B}_v$

$$(\mathrm{MSE})_{p,v}(x_0; h) = \hat{B}^2_{p,v}(x_0; h) + \hat{V}_{p,v}(x_0; h). \tag{3.14}$$

Eq. (3.14) would be used in the following manner to choose the optimal bandwidth:

$$\hat{h}_{p,v} = \arg\min_h \int \mathrm{MSE}_{p,v}(x, h)w(x)\mathrm{d}x, \tag{3.15}$$

where $w(.)$ is a given weighting function. The indicator function on the interval where the curve $g^{(v)}(.)$ is estimated is a common choice for such weighting function.

In most cases, a constant bandwidth will suffice; however, if the curve that is being estimated exhibits various degrees of smoothness at different locations then a variable bandwidth selector might be required. In this paper, we adopted a variable bandwidth adapter via splitting the data into small bins[4] and estimating h for those individual bins separately.

3.2. Implementation

After presenting the methodology of LML estimation, we want to provide the reader with a brief summary of the estimation procedure. As has been explained in the second section of this paper, our final goal is to use LML estimator in estimating the conditional expectations in the conditional CAPM. In other words, we want to estimate the conditional means of the portfolio returns (market and individual portfolio); therefore, the coefficient of interest would the first beta coefficient, the coefficient of constant term in the X matrix, in Eq. (3.1). Moreover, since $v = 0$, $p = 1$ would be a good choice for simplicity purposes. As suggested by Fan and Gijbels [10], the value of "a" would be chosen as 2 for the higher order fit that would be used to estimate the bias and variance. Below we summarize the estimation procedure.

Firstly, a 3rd order polynomial would be fitted using a multiplicative grid of bandwidth values,[5] and a pilot bandwidth would be chosen, via RSC that is given by Eq. (3.12), to estimate the bias and variance of a 1st order fit. Then, these bias and variance estimates would be used with the bandwidth grid one more time to estimate the MSE given by Eq. (3.14) in order to choose the optimal bandwidth. In the final stage, this optimal bandwidth is used to estimate the LML estimates.

[4] See Appendix B for details.
[5] See Appendix B for details.

One handicap of LML estimation is that, in multivariate estimation, the dimension of X matrix can be very large. As it will be explained in the following section, we use four conditioning variables as suggested by Wang [31]. In that case the 3rd order fit in LML estimation would imply the X matrix to have nineteen columns. Another major obstacle that we face is the issue of singularities encountered in implementing the bandwidth selection procedure in practice.

Two main approaches have been adopted in this paper to overcome these problems. Firstly, we adopt a project pursuit approach to form an index from conditioning variables. In order to achieve this, a weighting matrix has been constructed. The first row of this matrix had equal weights for each conditioning variable. Then, the ratio of each conditioning variable's standard deviation to the sum of all conditioning variables' standard deviations has been computed. Increasing fractions of this ratio have been added to the equal weights to form the remaining rows of this matrix. After forming this matrix, an index of conditioning variables has been computed from each weighting row of this matrix and the weighting row which produces the minimum sum of squared residuals from a fit of conditional expectation of the portfolio returns on these indexed conditioning variables has been chosen as the optimal weighted index. This has been repeated for all of the conditional expectations that are required for estimating conditional CAPM. This approach has eliminated the multivariate estimation problems at the cost of the information lost from forming an index of these conditioning variables.[6]

Secondly, as an alternative approach, we adopt a partially additive model. Basically, conditional expected values of portfolio returns have been estimated for each individual conditioning variable, and then an average of these conditional estimates is taken. This approach also prevented the need for multivariate estimation in LML framework.

In the estimation of the multivariate Nadaraya–Watson kernel estimator, the independent multivariate normal density function was adopted as the choice of kernel. The standardized values of the conditioning variables were used for rescaling purposes as suggested by Hardle [19]. Moreover, the theoretical bandwidth that minimizes the approximate Mean Integrated Squared Error was chosen as the optimal bandwidth: $h = cN^{-1/2k+1}$ where N stands for sample size, k stands for the number of conditioning variables and c is a constant. As shown by Hardle [19], c is defined to be: $c = [4/(2k+1)]^{1/(k+4)}$ for a multivariate standard normal kernel.

In conclusion, we used two alternative approaches to test the conditional CAPM and compared the performance of these approaches with the multivariate Nadaraya–Watson kernel estimation that has been used in Wang's [31] paper. The first of these estimators relies on the additive model approach that has been implemented via LML estimation. The second estimator adopts the index approach in a LML framework. Monte Carlo evidence by Kayahan and Stengos [24] suggests that estimators based on the LML framework have better small and medium sample properties over simple kernel estimators and as such they offer a more appealing nonparametric methodology to estimate the conditional CAPM model. In the next section we will present the data and the empirical results of our analysis.

4. Empirical results

In this section, the empirical results obtained from testing of conditional Sharpe–Lintner CAPM will be presented. The description of the data used in this study is given below.

Following Fama and French [7] we use as individual portfolios five size and book-to-market portfolios that have the size and BE/ME quantile combinations SZ1/BM1, SZ1/BM5, SZ3/BM3, SZ5/BM1 and SZ5/BM5, where SZ1/BM1 stands for the portfolio of stocks that fall into the smallest size and book to market equity quantile. Moreover, the market portfolio (RP) was chosen to be the excess return on the market that is given by the value-weight return on all NYSE, AMEX, and NASDAQ stocks (from CRSP) minus the one-month Treasury bill rate. The data of both the market and individual portfolios were taken directly from Kenneth French's website. The sample that was used in this study covers monthly portfolio returns from March 1947 to January 1996,[7] consisting of 587 observations. A detailed description of stock returns can be found in Kenneth French's website.

The choice of conditioning variables was identical to Wang's [31] paper for comparison purposes.[8] Four conditioning variables, namely the dividend price ratio (DPR), the default premium (DEF), one-month Treasury bill

[6] See Kayahan and Stengos [24] for a detailed description of the indexing procedure.

[7] The time period after 1996 includes a number of financial crises that would add to fatter tails in the distribution of the rates of return and therefore offer potential improvements using LML techniques over standard kernel methods. However, the non-availability of data for the conditioning variables over that period constrains us to only use the data for the 1947–1996 time period.

[8] The data for the conditioning variables were kindly provided by Wang.

Table 1
Summary statistics

	DPR	DEF	RTB	EWR
Mean	4.03	0.92	4.64	0.78
St.Dev	1.06	0.43	2.99	4.85
Skewness	0.74	1.45	0.98	−0.12
Kurtosis	2.68	5.04	4.01	6.92
Jarque Bera	56.1	307.8	118.5	377.3
Cross correlations—conditioning variables				
DPR	1.00	0.19	0.00	−0.03
DEF	0.19	1.00	0.64	0.11
RTB	0.00	0.64	1.00	−0.11
EWR	−0.03	0.11	−0.11	1.00

	SZ1/BM1	SZ1/BM5	SZ3/BM3	SZ5/BM1	SZ5/BM5	RP
Mean	0.72	1.53	1.19	0.99	1.24	0.64
St.Dev	7.04	5.60	4.62	4.43	4.74	4.12
Skewness	−0.19	0.02	−0.48	−0.17	0.01	−0.45
Kurtosis	4.78	7.08	5.82	5.01	3.81	5.28
Jarque Bera	80.77	407.4	216.9	102.2	16.2	146.5
Cross correlations—portfolio returns						
SZ1/BM1	1.00	0.87	0.82	0.67	0.65	0.78
SZ1/BM5	0.87	1.00	0.89	0.65	0.75	0.79
SZ3/BM3	0.82	0.89	1.00	0.79	0.78	0.91
SZ5/BM1	0.67	0.65	0.79	1.00	0.68	0.94
SZ5/BM5	0.65	0.75	0.78	0.68	1.00	0.82
RP	0.78	0.79	0.91	0.94	0.82	1.00

rate (RTB) and the excess return on the NYSE equally weighted portfolio (EWR), were chosen among a group of ten variables that included factors such as industry growth rate, inflation rate, short-end term structure slope, January dummy etc. The definitions of the conditioning variables that were used in this study are as follows.

DPR is the dividend yield, in percent, on the NYSE value-weighted index, measured as the sum of the previous 12 months' dividend payments divided by the level of index. DEF stands for the Baa-rated corporate bond yield minus that of Aaa-rated bond. RTB represents the one month T bill yield. Finally, EWR is the excess return on the NYSE equally-weighted index. The conditioning variables were lagged one-month behind the stock returns.

After providing the detailed description of the data summary statistics are given in Table 1.

As it can be seen from the kurtosis and highly significant Jarque-Bera statistics none of the variables are normally distributed. All of the conditioning variables, except EWR, are positively skewed. Moreover, the cross-correlations among conditioning variables are low except the RTB–DEF pair, giving support to the partially additive approach that is adopted in the LML framework. The portfolio returns also suffer from non-normality features and the cross correlations of portfolio returns and market portfolio returns (RP) are considerably higher as expected.

The results from chi-square test results of conditional CAPM and the properties of the pricing errors will be presented for the three estimators. The chi-square results obtained from the joint test of size and book-to-market portfolios are given in Table 2.

Table 2 shows that the conditional Sharpe–Lintner CAPM is rejected strongly for each and every one of the estimation methods. The significance tests of individual regressors imply that the major source of rejection originates from the highly significant intercepts in each estimator. This was also a conclusion of Wang [31]; however, different from his results none of the individual regressors except the intercept are found to be significant. In order to get a different perspective in the performance of our estimators, the descriptive statistics of the pricing errors are evaluated next. In detail, the biases, standard deviations and root mean squared errors of the pricing errors for each portfolio as well as the averages of these indicators for the pricing errors of all portfolios were computed. This has been done via two methods for robustness purposes, following Wang [31]. The first method is to get the fitted values from the Weighted Least Squares regression given by Eq. (2.8) as the estimator of the pricing errors

Table 2
Chi-square tests

		Joint test	Significance of individual regressors				
			Intercept	Def	Dpr	Rtb	Ewr
NWK (Multivariate)	χ^2-stat	48.81	28.57	5.08	2.98	7.29	3.48
	p-value	0.0003	0.000	0.406	0.702	0.200	0.627
LMLH (Additive)	χ^2-stat	57.811	31.78	7.09	3.93	8.11	3.73
	p-value	0.000	0.000	0.214	0.560	0.151	0.589
LMLH (Index)	χ^2-stat	56.849	19.22	5.42	9.87	7.22	5.33
	p-value	0.000	0.002	0.367	0.079	0.205	0.377

Table 3
Estimated pricing errors with WLS regression

	LMLH (Index)	NWK (Multivariate)	LMLH (Partially additive)
Mean($\varepsilon_{1,t}$)	−1.140	−0.202	−0.260
Mean($\varepsilon_{2,t}$)	−0.221	0.559	0.668
Mean($\varepsilon_{3,t}$)	−0.327	0.389	0.404
Mean($\varepsilon_{4,t}$)	−0.383	0.346	0.337
Mean($\varepsilon_{5,t}$)	−0.043	0.592	0.606
Std($\varepsilon_{1,t}$)	1.791	0.398	0.559
Std($\varepsilon_{2,t}$)	1.668	0.469	0.747
Std($\varepsilon_{3,t}$)	1.472	0.270	0.540
Std($\varepsilon_{4,t}$)	1.413	0.243	0.362
Std($\varepsilon_{5,t}$)	1.106	0.474	0.718
Rmse($\varepsilon_{1,t}$)	2.123	0.446	0.617
Rmse($\varepsilon_{2,t}$)	1.682	0.730	1.003
Rmse($\varepsilon_{3,t}$)	1.508	0.473	0.674
Rmse($\varepsilon_{4,t}$)	1.464	0.422	0.494
Rmse($\varepsilon_{5,t}$)	1.107	0.758	0.940
AAB	0.423	0.417	0.455
ASD	1.490	0.371	0.585
ARMSE	1.577	0.566	0.746

$$\hat{\varepsilon}_{i,t} = z_t' \hat{\delta}_i. \tag{4.1}$$

Secondly, the pricing errors were obtained from the description of CAPM:

$$\hat{\varepsilon}_{i,t} = \hat{g}_i(x_t) - \hat{g}_p(x_t)\hat{g}_{ip}(x_t)/\hat{g}_{pp}(x_t) \tag{4.2}$$

where the functions in Eq. (4.2) are the nonparametric estimates of variables in Eq. (2.7).

The three summary measures of the cross-section of five pricing error series for three alternative estimators are given in Table 3. The results of this analysis are similar to the chi-square test with one major difference. It can be seen from the table that the average absolute bias is more or less the same for the three nonparametric estimators. The major difference comes from the volatility of the pricing errors estimated by these methods. The multivariate NWK estimator has the smallest volatility on average for all pricing errors so it performs best. The additive LML approach performs marginally worse than the NWK estimator, but its performance is close to that. Of all three the LML index estimator performs the worst, since the volatility of the indexed LML estimator is by far the highest as manifested by its pricing errors. This result might be due to the fact that the index formed from conditioning variables is not adequate in terms of jointly reflecting the information provided by these conditioning variables. In order to have a robustness check, the summary measures of the pricing errors that are computed with the second approach should also be checked. The results from this approach are presented in Table 4.

The results obtained from this alternative method of pricing error calculations for robustness check purposes are suggesting that the additive LML outperforms the multivariate NWK, something that was not evident from the results of Table 3. One consistent result from the analysis of both alternative pricing error estimates is the poor performance

Table 4
Estimated pricing errors with robustness check method

	LMLH (Index)	NWK (Multivariate)	LMLH (Partially additive)
Mean($\varepsilon_{1,t}$)	−0.076	−0.117	−0.166
Mean($\varepsilon_{2,t}$)	0.824	0.726	0.752
Mean($\varepsilon_{3,t}$)	0.566	0.463	0.495
Mean($\varepsilon_{4,t}$)	0.329	0.272	0.298
Mean($\varepsilon_{5,t}$)	0.618	0.531	0.556
Std($\varepsilon_{1,t}$)	1.752	1.044	0.751
Std($\varepsilon_{2,t}$)	1.358	0.970	0.677
Std($\varepsilon_{3,t}$)	1.064	0.547	0.485
Std($\varepsilon_{4,t}$)	0.348	0.356	0.449
Std($\varepsilon_{5,t}$)	1.294	0.538	0.471
Rmse($\varepsilon_{1,t}$)	1.753	1.051	0.769
Rmse($\varepsilon_{2,t}$)	1.589	1.212	1.012
Rmse($\varepsilon_{3,t}$)	1.205	0.717	0.693
Rmse($\varepsilon_{4,t}$)	0.479	0.448	0.539
Rmse($\varepsilon_{5,t}$)	1.434	0.756	0.729
AAB	0.483	0.422	0.454
ASD	1.163	0.691	0.567
ARMSE	1.292	0.837	0.748

of the indexed LML estimator compared to the other estimators due to the presence of significant volatility in its estimates.

5. Conclusion

The conditional Sharpe–Lintner CAPM was tested with the nonparametric stochastic discount factor approach, following Wang [31], in this paper. Multivariate Nadaraya–Watson Kernel estimator and two alternative forms of Local Maximum Likelihood estimator were adopted to compare the performance gains from various nonparametric estimation techniques.

In the empirical section the results were in agreement in rejecting the conditional CAPM by every one of the three nonparametric estimators. We find that Wang's [31] findings are robust to the use of different nonparametric smoothers and they do not seem to come about by poor estimation at the extremes of the sample. The major source of rejection was observed to be coming from the intercept in the weighted least squares regression that seems to be a stylized fact in the history of empirical testing of CAPM.

The robustness of Wang's [31] results combined with the fact that a major source of CAPM's rejection arises from the significance of the intercept term in the conditional pricing errors regression implies the presence of unaccounted factors in the model. One possible reason for this outcome might be the failure of the stock market index to act as a sufficiently close proxy for the market portfolio. A possible venue for future research is to extend the analysis by incorporating a proxy for the returns to human capital into the model as suggested by Jagannathan and Wang [21] in order to obtain a more accurate index.

Acknowledgement

The second author would like to acknowledge financial support from the SSHRC of Canada.

Appendix A. Estimation procedure of variance–covariance matrix, Ω

The asymptotic results for nonparametric test in the context of conditional single beta models are presented in Wang [30]. Wang [31] presents the details of asymptotic variance–covariance matrix of $\hat{\delta}_N$ and its estimator, $\hat{\Omega}_N$, as follows:

Let $r_{t+1} = (r_{1,t+1} \dots r_{n,t+1})' \otimes z_t$ and $y_{t+1} = (x_t' z_t' r_{p,t+1} r_{t+1}')'$ where '$\otimes$' stands for the Kronecker operator. Moreover, denote $w_t = f(x_t) g_{pp}(x_t)$, $A = \iota_n \otimes E[w_t z_t z_t']$, and $\hat{A}_N = \iota_n \otimes N^{-1} \sum_{t=1}^{N} \hat{w}_t z_t z_t'$, where ι_n is the $n \times n$ identity matrix. Let $\delta = (\delta_1' \dots \delta_n')'$ with $\delta_i = [E(w_t z_t z')]^{-1} E(w_t z_t e_{i,t+1})$.

Define the following variables:

$$\gamma(y_{t+1}) = \eta(y_{t+1}) - [\iota_n \otimes a(y_{t+1})]\delta,$$
$$\eta(y_{t+1}) = f(x_t)[g_{pp}(x_t)r_{t+1} - g_p(x_t)r_{p,t+1}r_{t+1} + g_r(x_t)r_{p,t+1}^2 - g_{pr}(x_t)r_{p,t+1}]$$
$$a(y_{t+1}) = f(x_t)[g_{pp}(x_t)z_t z_t' - r_{p,t+1}^2 g_{zz}(x_t)]$$

where

$$g_r(x_t) = E(r_{t+1} \mid x_t), \qquad g_{pr}(x_t) = E(r_{p,t+1}r_{t+1} \mid x_t) \quad \text{and} \quad g_{zz}(x_t) = E(z_t z' \mid x_t).$$

Then, the variance–covariance matrix Ω is given by: $\Omega = A^{-1}\Gamma A^{-1}$ where Γ is given by: $\Gamma = \sum_{-\infty}^{\infty} \Gamma_j$ and $\Gamma_j = E[\gamma(y_{t+1})\gamma(y_{t+j+1})']$

In order to estimate the covariance matrix, $\gamma(y_{t+1})$ should be estimated via replacing the functions $f(x)$, $g_p(x)$, $g_{pp}(x)$, $g_r(x)$, $g_{pr}(x)$ and $g_{zz}(x)$ by their nonparametric estimators. The Local Maximum Likelihood method was used for estimating these functions nonparametrically except the $f(x)$, $g_r(x)$ and $g_{pr}(x)$ functions that are estimated with the multivariate Nadaraya–Watson kernel estimator, since we wanted the testing procedure to be consistent with Wang in general. In practice, $\gamma(y_{t+1})$ is estimated with sample equivalents of its components via:

$$\hat{\gamma}(y_{t+1}) = \hat{\eta}_N(y_{t+1}) - [\iota_n \otimes \hat{a}_N(y_{t+1})]\hat{\delta}_N,$$
$$\hat{\eta}_N(y_{t+1}) = \hat{f}(x_t)[\hat{g}_{pp}(x_t)r_{t+1} - \hat{g}_p(x_t)r_{p,t+1}r_{t+1} + \hat{g}_r(x_t)r_{p,t+1}^2 - \hat{g}_{pr}(x_t)r_{p,t+1}]$$
$$\hat{a}_N(y_{t+1}) = \hat{f}(x_t)[\hat{g}_{pp}(x_t)z_t z_t' - r_{p,t+1}^2 \hat{g}_{zz}(x_t)].$$

$\hat{\Gamma}_j$ can be shown to be a consistent estimator of Γ_j, and it is expressed as:

$$\hat{\Gamma}_j = N^{-1} \sum_{t=1}^{N-j} \hat{\gamma}_N(y_{t+1})\hat{\gamma}_N(y_{t+1})'.$$

The covariance estimator that is used by Wang [31] is also adopted in this paper:

$$\hat{\Omega}_N = \hat{A}_N^{-1}\hat{\Gamma}_o\hat{A}_N^{-1}.$$

Appendix B. Bandwidth grid and data bin selection for LML method

The variable bandwidth approach requires a grid of bandwidth values for the pilot and optimal bandwidth selection and it also requires the data to be split into a number of bins that depend on the sample size.

Fan and Gijbels [10] suggest the following data-dependent procedure to select the number of databins that the data will be divided into and the values for the bandwidth grid.

They suggest the formula $n_{\text{grid}} = [n/(10 \log n)]$, where n stands for the sample size, in order to determine the number of bins that the sample will be divided into. Then, after determining the number of grids for the data, we apply the following procedure to compute the grid values for the bandwidth.

Choose the first grid point via: $h_{\min} = (X_{(N)} - X_{(1)})/N$ where N stands for the number of observations in the databin, $X_{(N)}$ stands for the biggest independent variable value in the databin and $X_{(1)}$ is the smallest independent variable value in the databin. Similarly, compute the last grid point with $h_{\max} = (X_{(N)} - X_{(1)})/2$. Finally, obtain the other grid points with the following formula: $h_j = C^j h_{\min}$ where j stands for the jth value of the bandwidth grid and C stands for the factor that is used for inflating the h values. In practice, C was chosen to be 1.1 by Fan and Gijbels [10].

In summary, the number of databins would be chosen as the first step in practice. In the second step, $h_{\min}$ and $h_{\max}$ will be computed. Finally, $h_{\min}$ will be inflated until it reaches $h_{\max}$.

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